CS471 – Lesson 4 In-Class Worksheet Solutions

Search Problems & Problem Formulation

Guided In-Class Worksheet (Instructor-Led)

Objective: Walk through the formal definition of a search problem using a standard grid environment.

Scenario: A reconnaissance UAV must navigate a hostile sector represented by an $M \times N$ grid. The objective is to find a safe flight path from its launch point at $(0, 0)$ in the bottom left corner to the extraction point at (M, N) in the top right corner.

Part 1: Problem Formulation

Describe the problem using the formal terms of a search problem:

1. **States:**

   Solution

   Step 1: Identify what information is required to abstract the world.
   Step 2: The UAV's position is the only changing variable that matters for navigation.
   Answer: The set of all possible (x, y) coordinates within the $M \times N$ grid.

2. **Actions (and costs):**

   Solution

   Step 1: Determine the legal moves the agent can make from any given state.
   Step 2: Assign a mathematical penalty (cost) to these actions.
   Answer: Move North, South, East, or West. (Assuming uniform fuel consumption, Cost = 1 per move).

3. **Successor Function:**

   Solution

   Step 1: Define the mathematical function that takes the current state and an action, and returns the new state and cost.
   Answer: $f((x, y), \text{Action}) \rightarrow ((x', y'), \text{Cost})$. For example, $f((0, 0), \text{North}) \rightarrow ((0, 1), 1)$.

4. **Start State:**

Solution

Step 1: Identify the initial configuration of the agent.
Answer: The launch coordinate $(0, 0)$.

5. **Goal Test:**

Solution

Step 1: Define the condition that mathematically triggers the end of the search.
Answer: Does the current state (x, y) exactly equal the extraction point (M, N)?

Part 2: State Space Evaluation

6. **What is the size of the State Space for this problem?**

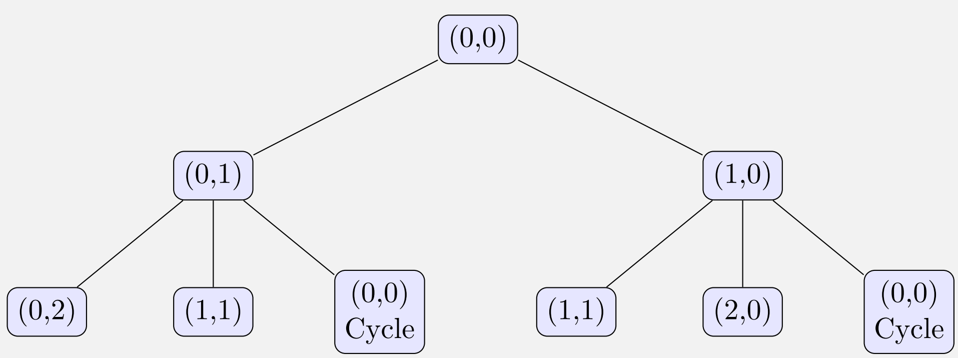
Solution

Step 1: Calculate the total number of unique, physically possible configurations in this abstracted world.
Step 2: The grid is M units wide and N units tall.
Answer: The size of the state space is $M \times N$ total states.

7. **Draw the first three layers of the Search Tree (Assume $M = 3, N = 3$):**

Solution

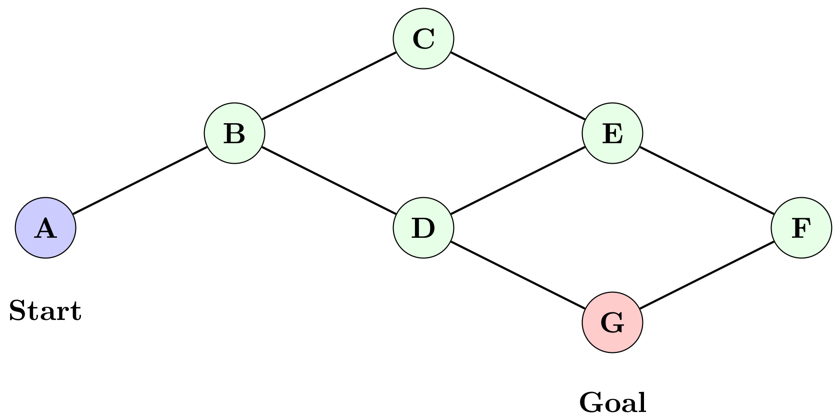
Step 1: Start at the root node (0,0).
Step 2: Expand the valid actions for the first layer (cannot move South or West due to grid boundaries).
Step 3: Expand the valid actions from the layer 1 nodes.
Answer:



Student Practice

Objective: Independently formulate a search problem based on a non-grid waypoint map.

Scenario: Now consider a path problem similar to a map. A ground convoy must navigate a secure road network. The map consists of specific tactical waypoints (A through G) connected by roads.



Problem Formulation

For the map, describe the core components of the search problem:

1. **States:**

   Solution

   Step 1: Identify the discrete locations the convoy can occupy.
   Answer: The set of waypoints: {A, B, C, D, E, F, G}.

2. **Actions (and costs):**

   Solution

   Step 1: Identify how the convoy transitions between states.
   Step 2: Note the lack of explicit distances on the map, implying uniform costs.
   Answer: Travel along an adjacent connecting road to a neighboring waypoint. (Cost = 1 per road segment).

3. **Successor Function:**

   Solution

   Step 1: Define the transition mapping.
   Answer: $f(\text{Current\_Waypoint}, \text{Travel\_Road}) \rightarrow (\text{Next\_Waypoint}, \text{Cost})$.

4. **Start State / Goal Test:**

Solution

Step 1: Select logical starting and ending points based on the graph structure. (Assuming left-to-right traverse).
Answer:
- **Start State:** Waypoint A
- **Goal Test:** Is Current_State == Waypoint G

Search Space Evaluation

5. **Draw the Search Tree:**
Trace the potential paths from Node A up to a depth of 3. Note any redundant paths (cycles).

Solution

Step 1: Set A as the root.
Step 2: Expand to A's only neighbor (B).
Step 3: Expand B's neighbors (C and D), excluding the immediate parent (A) to prevent trivial looping.
Step 4: Expand C and D's neighbors.
Answer: